Caffeine does not increase the risk of atrial fibrillation: a systematic review and meta-analysis of observational studies.

BACKGROUND: Atrial fibrillation (AF) is the most prevalent sustained arrhythmia, and risk factors are well established. Caffeine exposure has been associated with increased risk of AF, but heterogeneous data exist in the literature. OBJECTIVE: To evaluate the association between chronic exposure to caffeine and AF. DESIGN: Systematic review and meta-analysis of observational studies. DATA SOURCES: PubMed, CENTRAL, ISI Web of Knowledge and LILACS to December 2012. Reviews and references of retrieved articles were comprehensively searched. STUDY SELECTION: Two reviewers independently searched for studies and retrieved their characteristics and data estimates. DATA SYNTHESIS: Random-effects meta-analysis was performed, and pooled estimates were expressed as OR and 95% CI. Heterogeneity was assessed with the I(2) test. Subgroup analyses were conducted according to caffeine dose and source (coffee). RESULTS: Seven observational studies evaluating 115 993 individuals were included: six cohorts and one case-control study. Caffeine exposure was not associated with an increased risk of AF (OR 0.92, 95% CI 0.82 to 1.04, I(2)=72%). Pooled results from high-quality studies showed a 13% odds reduction in AF risk with lower heterogeneity (OR 0.87; 95% CI 0.80 to 0.94; I(2)=39%). Low-dose caffeine exposure showed OR 0.85 (95% CI 0.78 to 92, I(2)=0%) without significant differences in other dosage strata. Caffeine exposure based solely on coffee consumption also did not influence AF risk. CONCLUSIONS: Caffeine exposure is not associated with increased AF risk. Low-dose caffeine may have a protective effect.